Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Hydrolysis of $CH_3CH_2NO_2$ with $85\%H_2SO_4$ gives

(A) CH_3CH_2OH

(B) C_2H_6

(C) $CH_3CH = NOH$

(D) CH_3COOH

Q.2

Towards electrophilic substitution, the most reactive will be

(A) Nitrobenzene

(B) Aniline

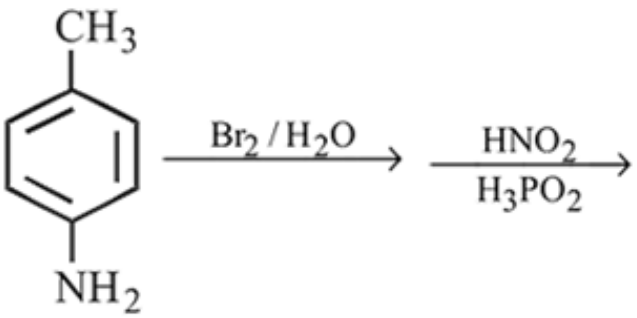
(C) Aniline hydrochloride

(D) N-Acetylaniline

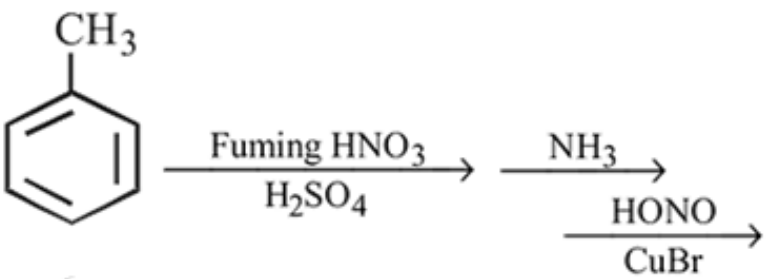
Q.3

3,5 -dibromotoluene can be best synthesised by

(A)

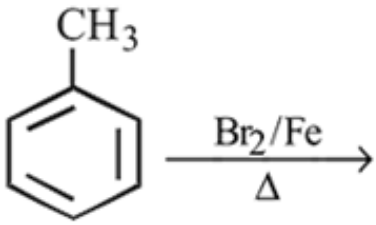


(B)

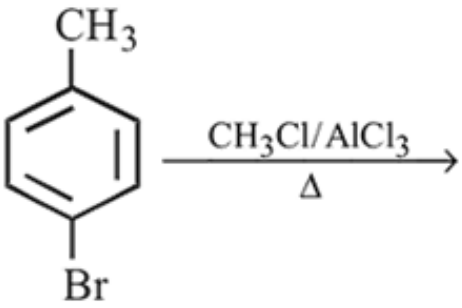


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(C)



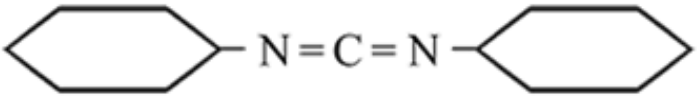
(D)



Q.4

Which of the following compounds is an enamine?

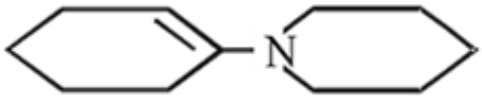
(A)



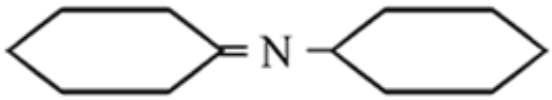
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(C)



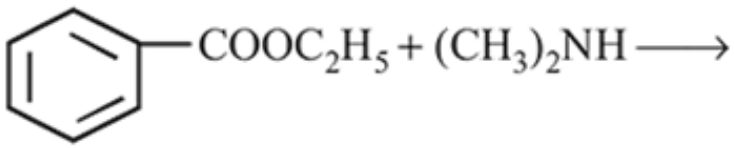
(D)



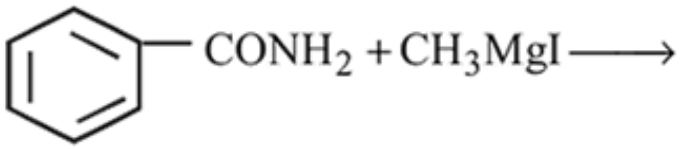
Q.5

Which of the following reactions will not give N, N- dimethyl benzamide?

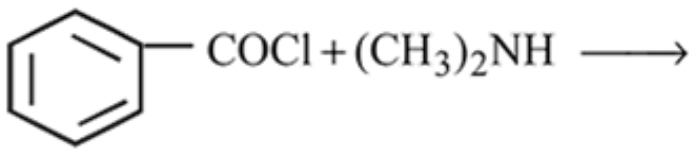
(A)



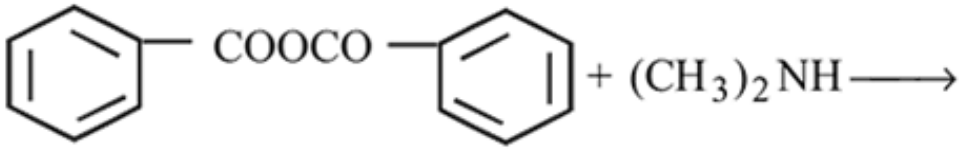
(B)



(C)

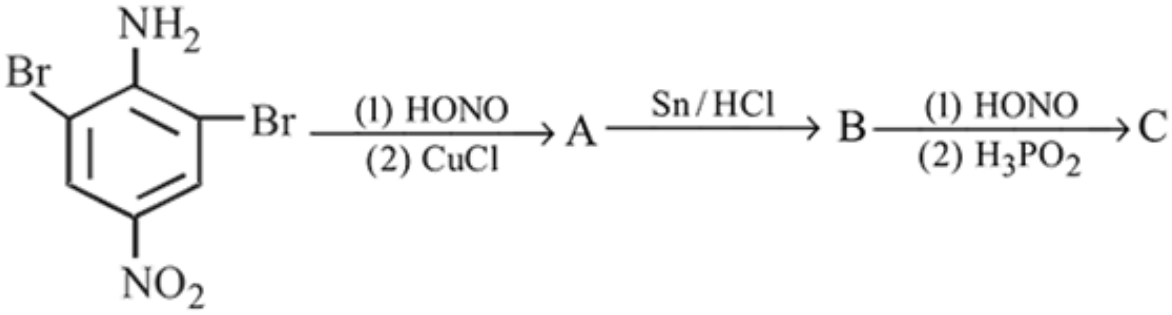


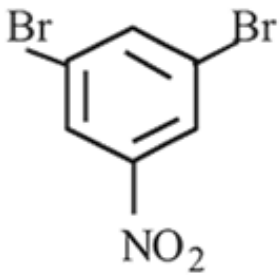
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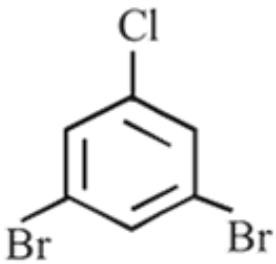
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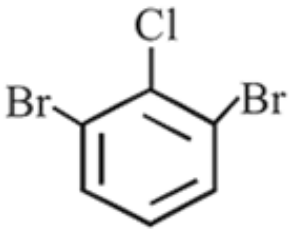
Q.6

The product $-(C)$ obtained in the following sequence of reactions is



(A) 

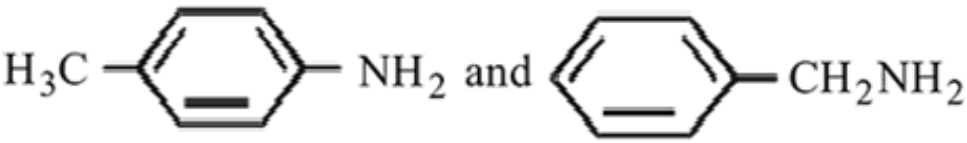
(B) 

(C) 

(D) None of these

Q.7

Which of the following reagents will be useful to distinguish between



?

(A) Dilute HCl

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(B) $C_6H_5SO_2$ and OH/H_2O

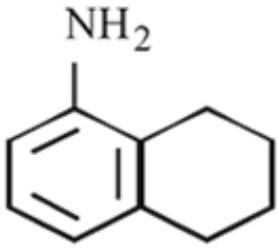
(C) HONO then β -naphthol

(D) $AgNO_3$ in H_2O

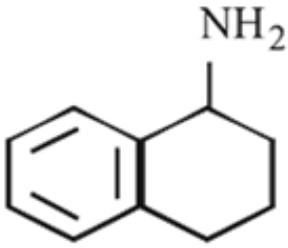
Q.8

Which of the following is the strongest base?

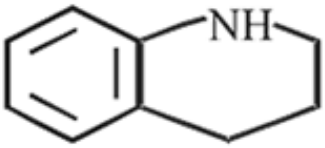
(A)



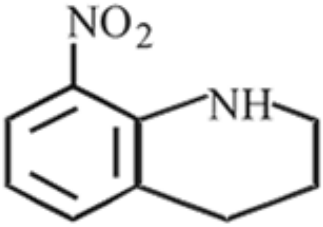
(B)



(C)



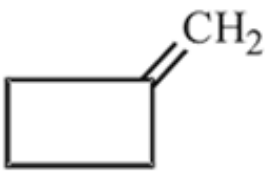
(D)



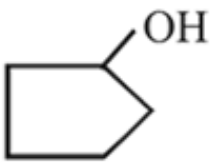
Q.9

The cyclobutyl methylamine with nitrous acid gives

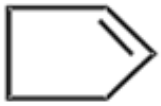
(A)



(B)



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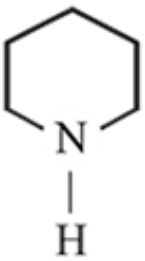
(C) 

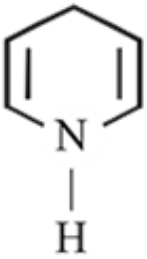
(D) All of these

Q.10

Which of the following amines will react with cyclohexanone to give enamine ?

(A) $(CH_3)_2NH$

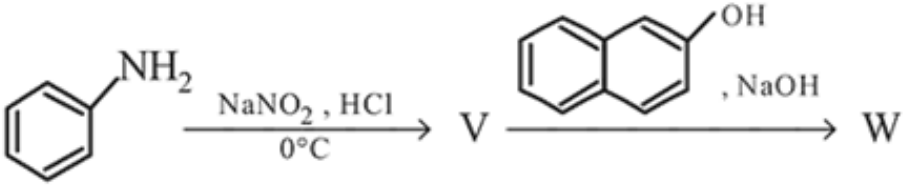
(B) 

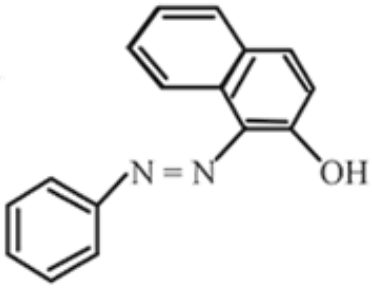
(C) 

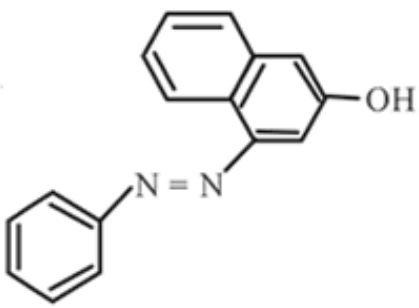
(D) All of these

Q.11

In the following reactions, the major product W is

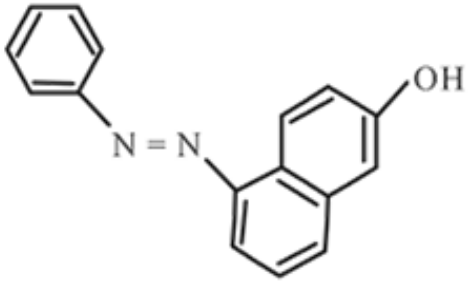


(A) 

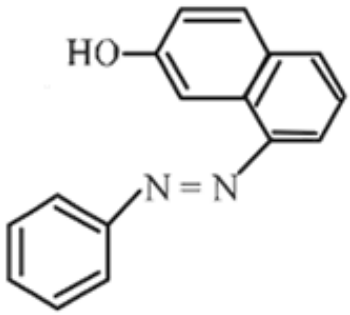
(B) 

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(C)



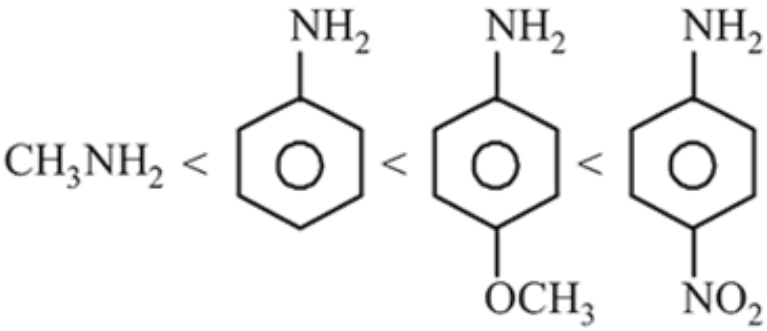
(D)



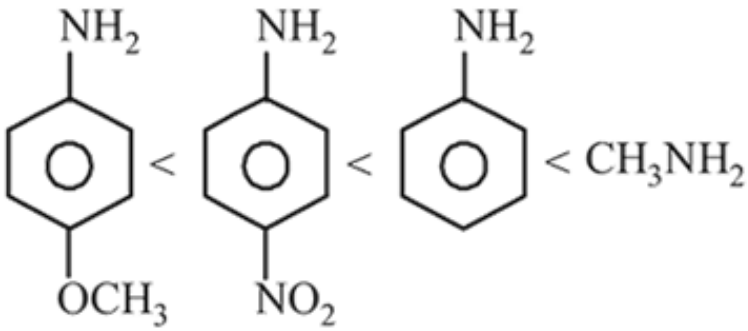
Q.12

Arrange the following amines in the order of increasing basicity.

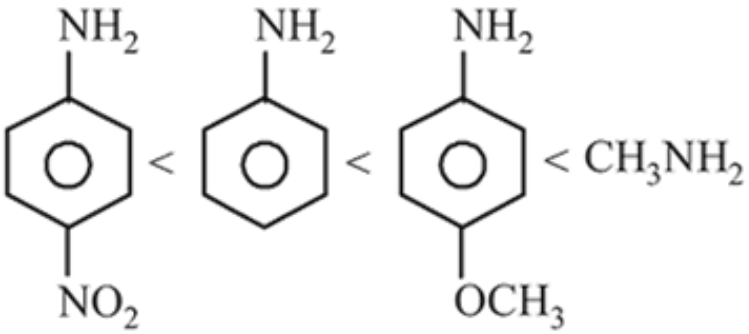
(A)



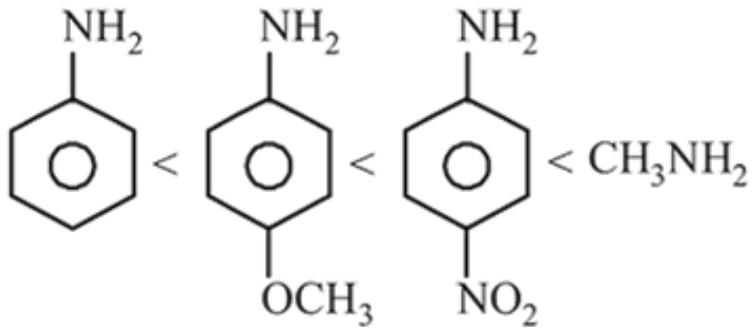
(B)



(C)



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(D) 

Q.13

The final product formed when Methyl amine is treated with $NaNO_2$ and HCl is:

(A) Diazomethane

(B) Methylalcohol

(C) Methylcyanide

(D) Nitromethane

Q.14

The order of basicity of amines in gaseous state is:

(A) $1^\circ > 2^\circ > 3^\circ > NH_3$

(B) $3^\circ > 2^\circ > NH_3 > 1^\circ$

(C) $3^\circ > 2^\circ > 1^\circ > NH_3$

(D) $NH_3 > 1^\circ > 2^\circ > 3^\circ$

Q.15

Carbylamine forms from aliphatic or aromatic primary amine via which of the following intermediates?

(A) Carbanion

(B) Carbene

(C) Carbocation

(D) Carbon radical

Q.16

The most basic compound among the following is

(A) Acetanilide

(B) Benzylamine

(C) p -Nitro aniline

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(D) Aniline

Q.17

Which of the following compounds gives dye test?

(A) Aniline

(B) Methylamine

(C) Diphenylamine

(D) Ethylamine

Q.18

An organic compound (A) on reduction gives compound (B). (B) on treatment with $CHCl_3$ and alcoholic KOH gives (C). (C) on catalytic reduction gives N-methylaniline. The compound A is

(A) Methylamine

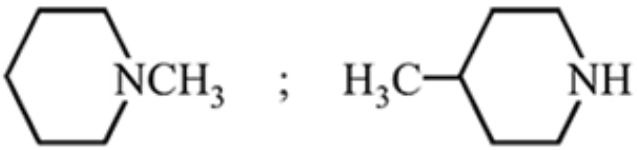
(B) Nitromethane

(C) Aniline

(D) Nitrobenzene

Q.19

Predict about the relative boiling point of the following two amines.



I II

(A) Boiling point of $I > II$

(B) Boiling point of $II > I$

(C) Both should have equal boiling points

(D) It can't be predicted

Q.20

Which one of the following is an oxidation product of a primary amine?

(A) A hydroxylamine

(B) A nitroso compound

(C) A nitro compound

(D) All of these

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Q.1 Answer:

CH_3COOH

Solution:

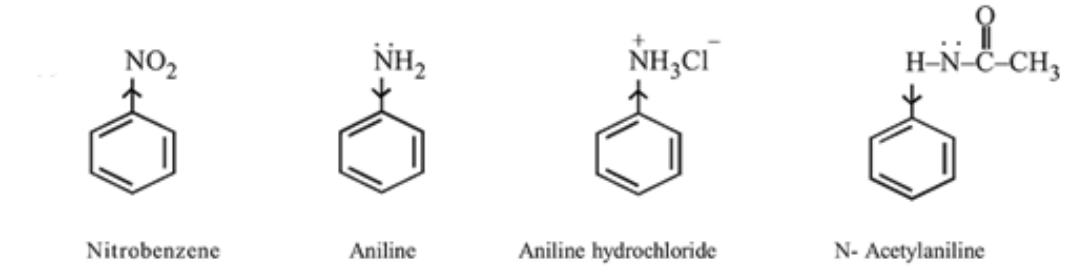
1° **Nitroalkanes on hydrolysis with boiling** $85\%H_2SO_4$ **give acids.**

$$CH_3CH_2NO_2 + H_2O + H_2SO_4 \xrightarrow{\text{heat}} CH_3COOH + [NH_3OH]^+ HSO_4^-$$

Q.2 Answer:

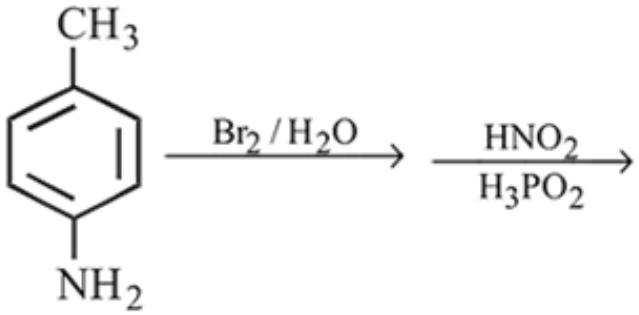
Aniline

Solution:

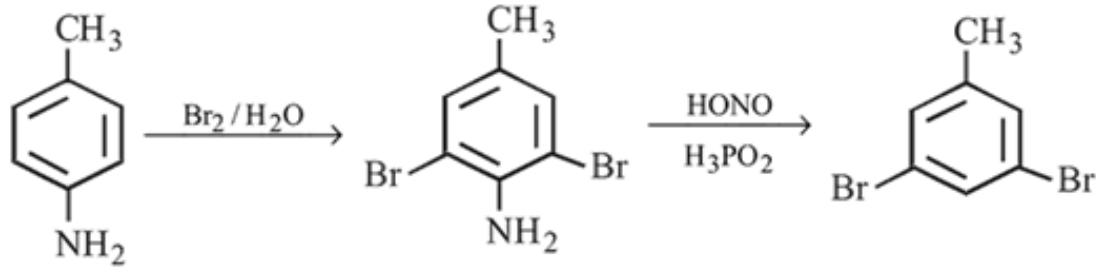


Nitrobenzene and aniline hydrochloride have electron- withdrawing $(-NO_2$ **and** $-NH_3^+)$ **groups, hence these will undergo electrophilic substitution with difficulty. Aniline and N-acetylaniline (acetanilide) have electron-releasing groups, however** $-NHCOCH_3$ **is less electron- releasing than** $-NH_2$ **due to delocalisation of lone pair of electron on** N **toward carbonyl group. Hence aniline (having** $-NH_2$ **) will undergo electrophilic substitution most easily.**

Q.3 Answer:

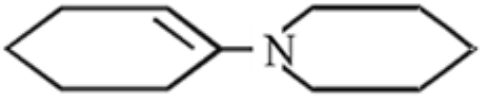


Solution:



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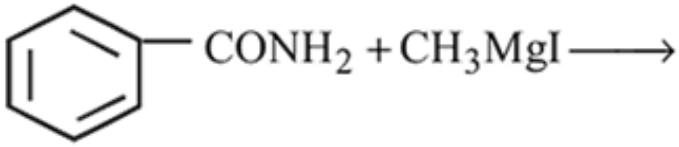
Q.4 Answer:



Solution:

Enamine (ene for carbon-carbon double bond and amine for amine group). $> C = C - N <$

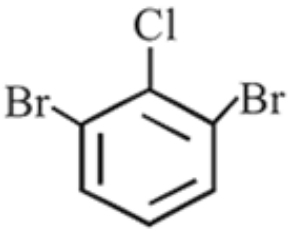
Q.5 Answer:



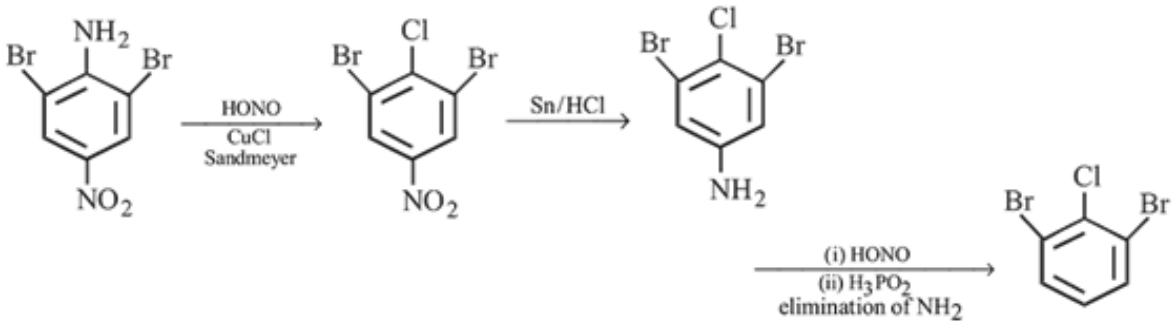
Solution:

(a) $C_6H_5COOC_2H_5 + (CH_3)_2\,NH \rightarrow C_6H_5CON\,(CH_3)_2 + C_2H_5OH$
(b) $C_6H_5CONH_2 + CH_3MgI \rightarrow C_6H_5CONHMgI + CH_4$
(c) $C_6H_5COCl + (CH_3)_2\,NH \rightarrow C_6H_5CON\,(CH_3)_2 + HCl$
(d) $C_6H_5CO.O \cdot COC_6H_5 + (CH_3)_2\,NH \rightarrow C_6H_5CON\,(CH_3)_2 + C_6H_5COOH$

Q.6 Answer:



Solution:



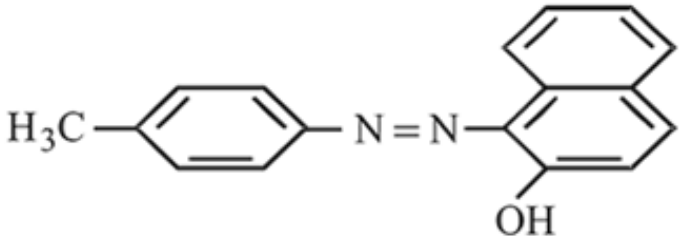
Q.7 Answer:

HONO then β -naphthol

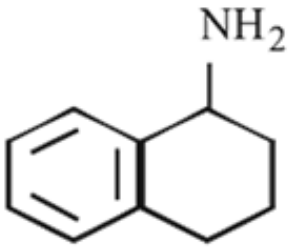
Solution:

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The first compound will give azo dye



Q.8 Answer:



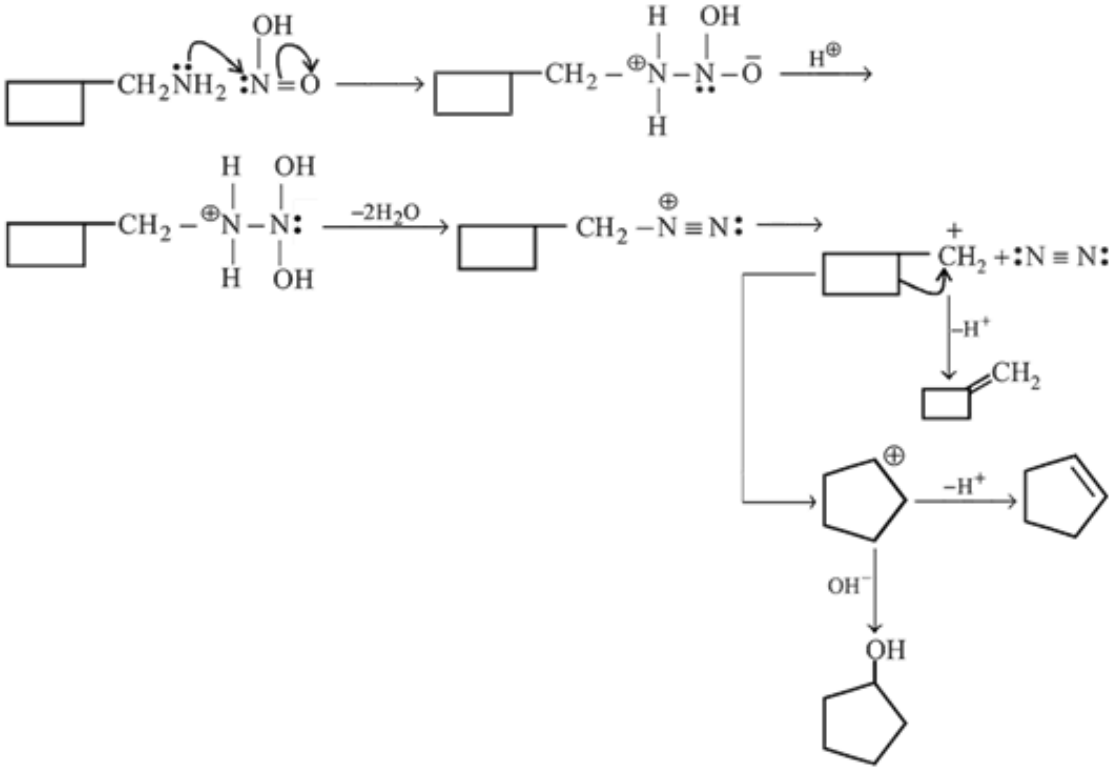
Solution:

Aliphatic amines are stronger base than aromatic amines. Further the order $2^\circ > 1^\circ > 3^\circ > \text{NH}$

Q.9 Answer:

All of these

Solution:



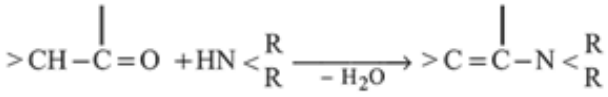
Q.10 Answer:

All of these

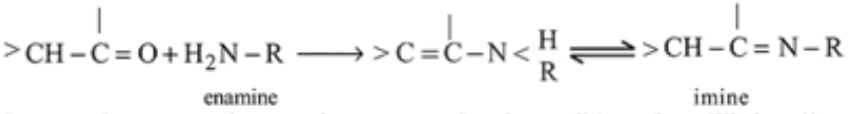
Solution:

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Carbonyl compounds containing at least one α -hydrogen atom react with a secondary amine to give enamine

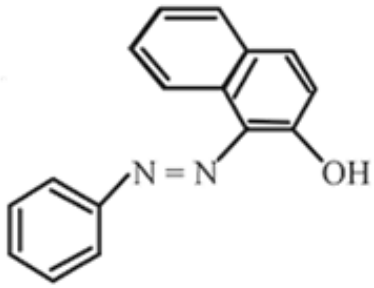


Primary amines also form enamines

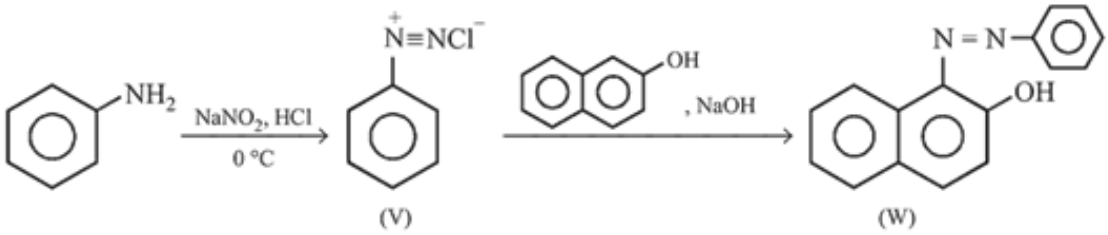


In second case enamine-amine tautomerism is possible and equilibrium lies completely on the imine side.

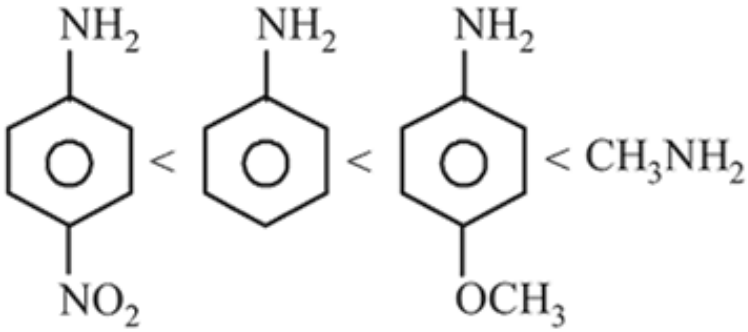
Q.11 Answer:



Solution:



Q.12 Answer:

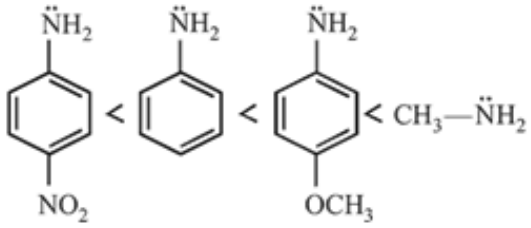


Solution:

Aliphatic amines are more basic than aromatic amines thus methylamine is most basic. Electron donating groups increase the basicity whereas electron withdrawing groups decrease the basicity of the aromatic amines. Thus p -methoxyaniline is more basic then aniline which is further more basic

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then p -nitroaniline.



Q.13 Answer:

Methylalcohol

Solution:

$$CH_3NH_2 + HNO_2 \longrightarrow CH_3OH + N_2 + H_2O$$

Q.14 Answer:

$3^\circ > 2^\circ > 1^\circ > NH_3$

Solution:

The correct order of relative basicity of amines in the gas phase is $3^\circ > 2^\circ > 1^\circ > NH_3$
The alkyl group releases electron and thus, tends to disperse the positive charge of the alkyl ammonium ion and therefore stabilises it. Since, NH_4^+ (from NH_3) has no such alkyl group, it is not stabilised to such an extent as alkylammonium ion.

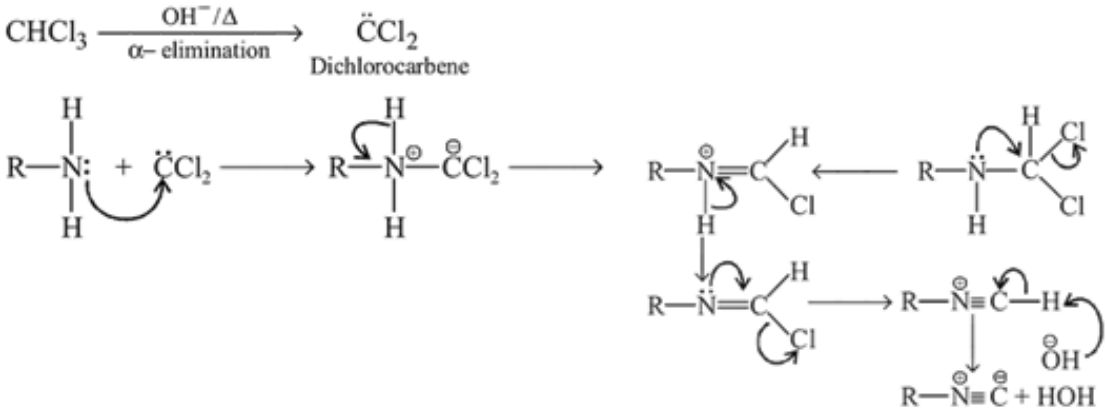
Q.15 Answer:

Carbene

Solution:

$$RNH_2 + CHCl_3 + 3KOH \longrightarrow RNC + 3KCl + 3H_2O$$

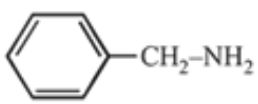
Mechanism



Q.16 Answer:

Benzylamine

Solution:



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compound is most basic due to presence of localized lone pair of electron on nitrogen atom while other compounds have delocalized lone pair of electron.

Q.17 Answer:

Aniline

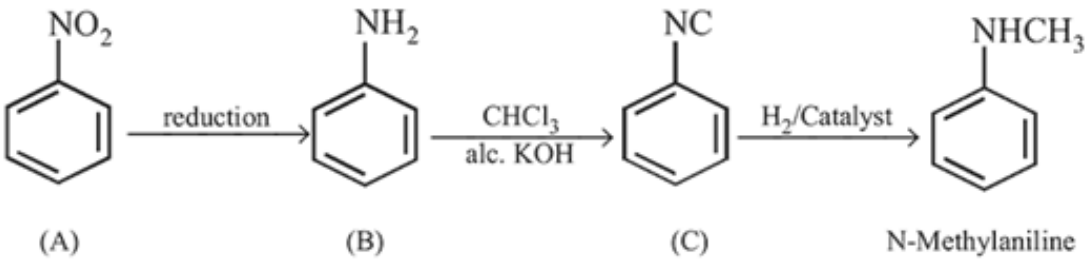
Solution:

Only 1° aromatic amines undergo coupling reactions to form a dye

Q.18 Answer:

Nitrobenzene

Solution:



Q.19 Answer:

Boiling point of II $>$ I

Solution:

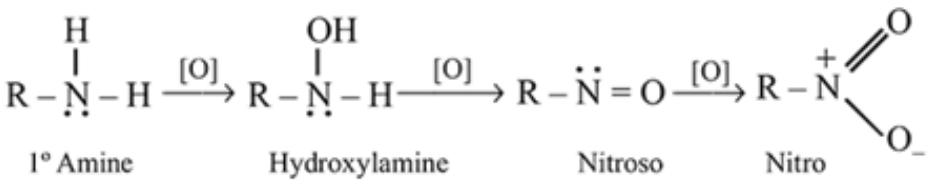
In II, intermolecular H-bonding between similar molecules is possible, while this is not so in I

Q.20 Answer:

All of these

Solution:

All of the three are oxidation products of a 1° amine.



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